

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```

1  #include <stdio.h>
2
3  void swap(int* a, int* b) {
4      int t = *a;
5      *a = *b;
6      *b = t;
7  }
8
9  int partition(int arr[], int low, int high) {
10     int pivot = arr[high];
11     int i = (low - 1);
12
13     for (int j = low; j <= high - 1; j++) {
14         if (arr[j] < pivot) {
15             i++;
16             swap(&arr[i], &arr[j]);
17         }
18     }
19     swap(&arr[i + 1], &arr[high]);
20     return (i + 1);
21 }
22
23 void quickSort(int arr[], int low, int high) {
24     if (low < high) {
25         int pi = partition(arr, low, high);
26         quickSort(arr, low, pi - 1);
27         quickSort(arr, pi + 1, high);
28     }
29 }
30
31 int main() {
32     int n;
33     if (scanf("%d", &n) != 1) return 0;
34
35     int arr[n];
36     for (int i = 0; i < n; i++) {
37         if (scanf("%d", &arr[i]) != 1) return 0;
38     }
39
40     quickSort(arr, 0, n - 1);
41
42     for (int i = 0; i < n; i++) {
43         printf("%d", arr[i]);
44         if (i < n - 1) {
45             printf(" ");
46         }
47     }
48     printf("\n");
49
50     return 0;
51 }
52

```

	Input	Expected	Got	
✓	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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